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Voice Commands With The Arduino Speech Recognition Engine

Control your device with voice commands using the Arduino Speech Recognition Engine

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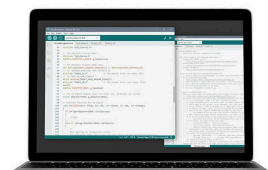
What Is Speech Recognition

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What Is Speech Recognition

Speech recognition is a technology field that captures, interprets, and computes a voice to transform it into text (TTS). Once the voice has been transformed into text, it can be applied to different applications, from speech dictation, to command-voice controllers, health monitoring, robotics and artificial intelligence or accessibility, among many others.

The Arduino Speech Recognition Engine

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The Arduino Speech Recognition Engine

The Arduino Speech Recognition Engine, which is powered by Cyberon, is one of the best platforms available on the market to perform speech recognition on embedded microcontrollers. The full compatibility with multiple Arduino boards and with the Arduino IDE makes it very flexible, efficient, and easy to use, both in hobby projects and in professional products.

The Arduino Speech Recognition Engine can be further customized based on the license type, check the [Licensing section](#) of this tutorial to learn more.

Overview

This tutorial shows how to use the Arduino Speech Recognition Engine to configure multiple voice commands and perform custom tasks based on the recognized commands. To do that, it is necessary to perform a series of steps to register your board and activate your trial and free-of-charge license. Afterward, you will test the demo sketch and learn how to create your own custom voice commands.

In case you are interested in unlocking the full potential of the tool, the tutorial will teach you how to acquire a paid license to unblock the free license limitations.

Goals

Set the voice command triggers

Get your board serial number and the required files to set up the library

Get the license and use the demo sketch

Create a new project with your own custom voice commands

Learn how to acquire and use a paid license in case you want to remove the free license limitations

Required Hardware and Software

To use the Arduino Speech Recognition Engine, you will need one of the following boards:

Arduino Portenta H7 (any variant) + Portenta Vision Shield (**LoRa** or **Ethernet**)

Arduino Nicla Vision

Arduino Nano 33 BLE Sense Rev 1

Arduino Nano 33 BLE Sense Rev 2

Arduino Nano RP2040

In the case of the Portenta H7, remember that is always possible to add an external microphone as additional hardware, instead of using the Portenta Vision Shield.

Additionally, you will need the following hardware and software:

USB-C® cable (x1)

Arduino IDE 1.8.10+, Arduino IDE 2.0+, Arduino Cloud Editor, or **Arduino-cli**

The Arduino Speech Recognition Engine is a solution powered by Cyberon that requires a series of additional steps to unleash its potential. This tutorial will explain how to:

- Get the Serial Number of your Arduino device

- Install the library on your IDE

- Test the free demo sketch

In case you would like to extend the engine functionalities, you will have to purchase [a voucher on the Arduino Store](#) and follow the next steps:

- Fill in the required information on Cyberon's website

- Get the required files and activate your license

Setup

Setup the Library

There are three libraries, you will need to install one or another depending on which board you are using:

Portenta H7 Family:

Cyberon_DSpotterSDK_Maker_
PortentaH7

Nicla Vision:

Cyberon_DSpotterSDK_Maker_
NiclaVision

Nano 33 BLE Sense (Rev1 & Rev2):

Cyberon_DSpotterSDK_Maker_
33BLE

Nano RP2040:

Cyberon_DSpotterSDK_Maker_
RP2040

Inside each of the libraries and under the folder "extra", you will find additional

 documentation made by Cyberon. Check them out in case you need more information

Go to the Library Manager, search for the library that you need for your board and install it.

In case you need more instructions about how to install libraries, read this [guide](#).

Get The Serial Number

To use the Arduino Speech Recognition Engine, you will need a free trial license or paid license. In any of the cases, the serial number of the board that you are using is necessary to activate the license.

To get your board's serial number, and once you have the library downloaded, navigate to **File > Examples > Cyberon_DSspotterSDK > GetSerialNumber**.

Connect your board to the computer, upload the sketch to it and, once is done, open the **Serial Monitor** to see your device's Serial Number.

On the Arduino IDE 1.6.x or previous versions, you can also find the serial number as

 follow: select the board's serial port and click on `tools > Get Board Info`, you will see the "SN" number, save it for later.


```
1  #include <stdint.h>
2
3  const uint32_t g_lp
4      0x00000000, 0x0
5      0x00000000, 0x0
6      0x00000000, 0x0
7      0x00000000, 0x0
8      0x00000000, 0x0
9      0x00000000, 0x0
10     };
```



Remember to replace the `g_lpdwLicense` values with your license ones.

Now switch back to the `VoiceRecognition` tab and upload the sketch.

You can now open the Serial Monitor where you can see the available commands that the CyberonSDK recognizes.



Serial Monitor Trigger and Command list

Now it is time to test the engine. Feel free to say out loud the commands and see how it recognizes your voice!

Please note that the sketch and license that you are using are a "demo" version with the following limitations:

- A maximum of 50 recognitions per each board start

- A 20 seconds delay before each trigger recognition

Customized Commands

To expand the features and human interaction of your projects, you can create your own voice commands with the Arduino Speech Recognition Engine. In this section, you will learn about how to use the [Cyberon Model Configuration](#) webpage to create a new project with custom voice commands.



These steps are compatible with the trial license, used in this tutorial, and the paid license

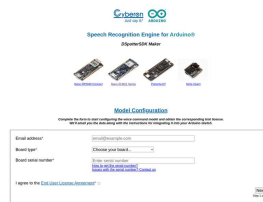
Go to [Cyberon Model Configuration](#) and fill in the required fields:

E-mail address

Board model

Serial Number of your board

EULA Agreement - Please, read it carefully



Cyberon Model Configuration

Click next, you will see a new page to:

Create a new project

Import an existing project

Create a New Project



Warning: each project is bound to a single Arduino board with the declared Serial Number. If you would like to use the Arduino Speech Recognition Engine on another Arduino board, you need to create a new project from scratch and assign it to the new Serial Number.

To create a new project first you need to select the desired language for the speech recognition. Once is set, click **Create**.



Cyberon, New Project

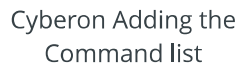
Now you need to configure the following steps to create a new trigger:

Create the **Input Trigger word**, for example "Hey Arduino". The **Input Trigger word** will trigger the device to let the board know that you are going to say a command after that.

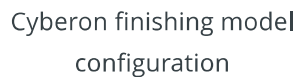


Cyberon Adding the Input trigger

Add the **Command** list. These commands will be used to do tasks on your sketch. For example, if you have a command which is **share**, later you will be able to get that command and proceed with some job inside the sketch easily.



On the next page you will see all the configurations already set. Check it out to confirm that everything is correct. In case something is wrong, click on **Back** to fix it.



i Warning: when a project is created and confirmed, it cannot be further modified.

On the IDE, open the example **File > Examples > Cyberon_DSspotterSDK > VoiceRecognition** and click **File > Save As...** and type a name for your sketch.

Once it is saved, open your File Explorer, and navigate to your sketch path.

On your sketch directory, paste the files you have got in your e-mail inbox before:

CybLicense_<id>.h

Model_<id>.h

Now to implement the **Input Trigger Command** and the **Command List** open the sketch and change the following headers with the downloaded ones:

```
1  ...
2
3  #include "CybLicense.h"
4
5  ...
6
7  #include "Model_L1.h" ->
8
9  ...
10
11 #include "Model_L0.h" ->
```

At this point, the project is set to be used. Upload the sketch and open the Serial Monitor. Now you can test your new **Input Trigger Word** and the **Command** list that you have created. Pronounce the new trigger words out loud, you will see the recognized words on the **Serial Monitor**.

Modify the Example Codes

If you want to execute custom actions with the **trigger** and **commands** you defined, you can do it inside the `VRCallback` function.

First you need to know your trigger and commands IDs. You can find them listed in your Serial Monitor in

Keyword	ID
Hey Arduino	100
read	10000
upload	10001
share	10002

Here is the `VRCallback` function modified so you can easily give functionality to the commands detection:

```
1 // Callback function for
2 void VRCallback(int nFlag)
3 {
4     if (nFlag==DSpotterSL)
5     {
6         //ToDo
7     }
8     else if (nFlag==DSpot)
9     {
10        /*
11         When getting an r
12         the following inc
13         nID          Th
14         nScore       nS
15         nSG          Th
16         nEnergy      nE
17         nEnergy      Th
18         */
19         //ToDo
20         switch(nID)
21         {
22             case 100:
23                 Serial.pri
24                 // Add your
25                 break;
26             case 10000:
27                 // Add your
28                 break;
```

We added a `switch...case` that evaluates the `nID` variable and compares it with the different IDs. Inside each case add your custom code to be executed with the trigger or commands

```
1  ...
2      switch(nID)
3      {
4          case 100:
5              Serial.print
6              // Add your
7              break;
8          case 10000:
9              Serial.print
10             // Add your
11             break;
12          case 10001:
13              Serial.print
14              // Add your
15              break;
16          case 10002:
17              Serial.print
18              // Add your
19              break;
20          default:
21              break;
22      }
23  ...
```

Unlock Limitations (License)

In case you want to further customize your model or add additional commands or trigger words, we have the right licenses for you:

Speech Recognition license.

Vouchers available on the [Arduino Store](#).

Pro License. Feel free to [contact us](#) to unlock all the limitations.

There are different licensing options to fit the needs of your project. Read more in the [Licensing section \(Voucher codes\)](#). Note that you need to have an already existing project to use a paid license. Check the [previous section](#) to know more.

Browse to [Cyberon's Licensed Project Configuration](#).

Fill in the required fields:

E-mail address

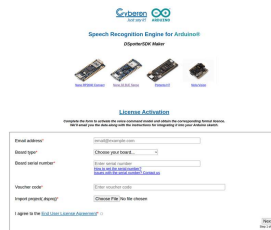
Board

Serial Number of your board

Voucher code

Import project: the file that you received in your e-mail inbox during the creation of the project.

EULA Agreement - Please read it carefully



Cyberon Model
Configuration with a
voucher code

Click next, review your project options and press continue.

You will get a new e-mail with the new License and Model headers.



Warning: in case you are using a Speech Recognition License (not Pro), please note that deployed models are not customizable. You can always purchase an additional voucher on [Arduino Store](#) to generate a new model for the same hardware.

Open the sketch you have duplicated in the [Create New Project](#) section.

Repeat the replacement of

```

1  ...
2
3  #include "CybLicense_<id>.h"
4
5  ...
6
7  #include "Model_<id>.h"
8
9  ...
10
11 #include "CybLicense_<id>.h"

```

Upload the sketch to your board, and you will have unlocked the limitations of the trial version.

Licensing

Here you can know more about the different licenses and limitations of each device. Remember that the code that you get for your license is also called a **voucher code**.

License specs			
	Speech Recognition Free Trial	Speech Recognition License	Pro License
n. of devices	1	1	unlimited
n. of triggers	1	1	unlimited
n. of commands	30 max	30 max	unlimited
max. delay	10	unlimited	unlimited
delay to trigger model*	20s	no	no

License options for the
Arduino Speech
Recognition Engine

Free Demo

Compatible with the
CDSpotterSDK library

Number of SET: x1

Number of TRIGGERS: x1

Number of COMMANDS: 4

Configure TRIGGER: Fixed (not
configurable)

Configure COMMANDS: Fixed
(not configurable)

Delay in Trigger mode: no

Hardware binding: no

Speech Recognition Free Trial

Once the number of recognition is reached the model will stop working and you will need to manually reboot the target.

Compatible with the
CDSpotterSDK library

Number of SET: x1

Number of TRIGGERS: x1

Number of COMMANDS: 20
maximum

Configure TRIGGER:
Configurable

Configure COMMANDS:
Configurable

Recognition times: x50

Delay in Trigger mode: 20
seconds [1]

Hardware binding: Yes

[1] Please note that the delay in Trigger mode means a 20 seconds delay between entering the trigger mode and the recognition of the wake-up word.

Speech Recognition License

Compatible with the
CDSpotterSDK library

Number of SET: x1

Number of TRIGGERS: x1

Number of COMMANDS: 20
maximum

Configure TRIGGER:
Configurable

Configure COMMANDS:
Configurable

Recognition times: Unlimited

Delay in Trigger mode: No
delay

PRO

Compatible with the
CDSpotterSDK PRO library

Number of SET: Unlimited
(depends on the hardware)

Number of TRIGGERS:
Unlimited (depends on the
hardware)

Number of COMMANDS:
Unlimited (depends on the
hardware)

Configure TRIGGER:
Configurable

Configure COMMANDS:
Configurable

Recognition times: Unlimited

Delay in Trigger mode: No
delay

Hardware binding: Yes

Next Steps

After getting the demo sketch working, we encourage you to start implementing this on your own project to unleash the full potential of the Arduino Speech Recognition Engine.

In case you are looking for more information, you can check out Cyberon's documentation:

Inside each of the libraries and under the folder "extra", you will find additional documentation made by Cyberon, as well as the `Readme.md` file that contains the requirements and the steps shown in this tutorial

[Cyberon's youtube channel](#) contains a series of step-by-step tutorials that can help you to know more about how to use the Arduino Speech

You can also consult
[Cyberon's official Maker User Guide](#)

Conclusion

In this tutorial, you have learned what is the Arduino Speech Recognition Engine, the different licenses and limitations, and how to acquire one. Once you have your license, you learned how to test it with the free demo sketch and how to create a project with your own custom voice triggers. Eventually, you learned how to upgrade a free already-made project to be used with a more advanced license.

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